Virtual Spatial Transcriptomics from H&E Histopathology: A Leakage-Aware Multimodal Deep Learning Study on Human Skin

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***Abstract*—**Spatial transcriptomics (ST) records gene expression together with tissue position, yet it remains costly and is available for relatively few cohorts, whereas haematoxylin and eosin (H&E) staining is routine and inexpensive. Recent foundation and generative models learn from very large collections of image–expression pairs, but small and transparent baselines evaluated under strict protocols are still useful for understanding how much expression morphology alone can explain. This paper presents a multimodal prototype that predicts spot-level expression of a fixed 56-gene panel from H&E patches of human skin (GSE197023: 19 Visium sections, 13,742 spots, atopic dermatitis and healthy controls). Each spot is described by frozen ResNet18 features of its own patch, the mean features of its six nearest neighbours and its standardised position within the section. Six models, from a training-mean baseline to a multimodal fusion network, are compared with sample-grouped 5-fold cross-validation, so that every test section is completely unseen during training. The fusion network performs best: MAE 0.7246 against 0.8251 for the baseline, mean R^2 0.0842, positive R^2 for 48 of 56 genes and mean within-sample Pearson correlation 0.2158. Epidermal-barrier and dermal-stromal genes are recovered best inside a section, whereas sparse immune genes remain largely unpredictable. All genes are reported, including the eight with negative R^2. The study is exploratory and is not a clinical or diagnostic system.

***Index Terms*—Spatial transcriptomics, histopathology, H&E, gene expression prediction, multimodal deep learning, atopic dermatitis.**

I. INTRODUCTION

Linking tissue morphology with molecular state has become a central goal of computational pathology, and spatial transcriptomics is the technology that makes the link measurable: platforms such as 10x Visium capture transcript counts at thousands of spatially registered spots on a single tissue section [1], [2]. The difficulty is practical. Sequencing is expensive, protocols evolve quickly and standards are scarce, which has kept most computational work confined to narrow tasks and small cohorts [3]. An H&E slide, by contrast, is produced for almost every biopsy. If expression could be estimated from the image alone, molecular maps could be obtained at a fraction of the cost. This idea is often called virtual spatial transcriptomics.

The first deep-learning attempts regressed expression directly from image patches of breast-tumour sections [4]. Later work embedded images and expression in a shared space with contrastive learning and imputed the expression of a query patch from a reference set [5]. More recently, large pretrained models have been trained on millions of image–transcriptome pairs [1], on whole slides with a geometry-aware transformer [2], or with flow matching [6]. Benchmarking of eleven methods [7] confirms that expression can be predicted from histology, but it also points to open issues in generalisability and translational use. Many of these systems depend on corpora and computing budgets that are out of reach for a small study, and it is less clear how a simple, fully inspectable pipeline behaves when whole tissue sections are held out.

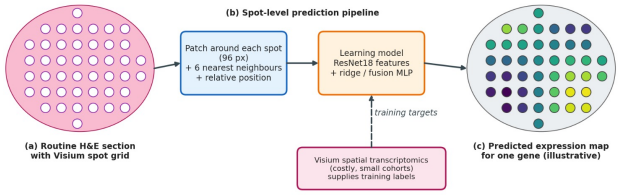


Fig. 1. Overview of virtual spatial transcriptomics. (a) A routine H&E section carries a grid of Visium spots. (b) Patches, neighbouring patches and relative position are passed to a learning model trained with measured Visium expression as targets. (c) The model outputs a spot-level expression map for each gene. Schematic drawn by the authors; task formulation follows [1].

In this work we study human skin. Healthy skin and atopic dermatitis (AD) share the same layered architecture but differ in barrier and immune programmes, so the data form a demanding test of whether morphology encodes expression. The study is intentionally modest in scope. It is inspired by, but does not reproduce, the foundation models of [1] and [2]; no foundation model is trained here, and the target is a fixed 56-gene panel rather than the full transcriptome. The main contributions are as follows: (1) a leakage-aware pipeline that converts 19 Visium sections (13,742 spots) into patch, neighbourhood and position features and predicts a 56-gene panel, with every scaling step fitted on training folds

only; (2) a comparison of six models under sample-grouped cross-validation, with R^2 reported against the training-mean reference; (3) complete per-gene reporting, including genes for which no model beats the mean; and (4) an analysis showing that structural genes are recovered inside a section while inflammatory genes are mainly separated between sections.

The rest of the paper is organised as follows. Section II reviews related work. Section III describes the dataset. Section IV presents the methodology. Section V reports results and observations. Section VI discusses limitations, future work and concludes.

II. RELATED WORK

This section reviews work on predicting spatial gene expression from histology and groups it into four lines: supervised and contrastive predictors, large pretrained models, pathology encoders, and benchmarks. Table I summarises the studies.

A. Supervised and Contrastive Predictors

ST-Net [4] showed that a convolutional network can learn a mapping from breast-tumour morphology to spatial expression. BLEEP [5] replaced direct regression with a bi-modal contrastive embedding, so that the expression of a query patch is imputed from the most similar patches of a reference dataset; it was demonstrated on human liver Visium data. These methods establish the task, but both were demonstrated on a narrow range of tissue, so their behaviour on a new organ or disease is not settled by the original reports.

B. Large Pretrained and Generative Models

OmiCLIP [1] pairs histology images with transcriptomes written as sequences of top-expressed gene symbols, trains a CLIP-style dual encoder on about 2.2 million pairs from 32 organs, and exposes gene-expression prediction as one of several downstream functions. STPath [2] is a generative model pretrained on whole-slide images with ST profiles; it integrates image, expression, organ and technology information in a geometry-aware transformer and predicts many thousands of genes without task-specific fine-tuning. STFlow [6] formulates prediction as flow matching over whole slides, and SPADE [9] aligns slide patches and expression through a mixture of data experts trained contrastively on HEST-1k [3]. The common trend is scale: more slides, more organs and more genes. The price is dependence on very large curated corpora such as HEST-1k, which pools 1,229 profiles from 153 cohorts [3], and on substantial computing resources.

C. Pathology Encoders

Most recent predictors do not learn image features from scratch. They reuse encoders pretrained on histology, for example the self-supervised vision transformer UNI trained on more than 100,000 whole slides [10] or the vision-language model CONCH trained on 1.17 million image–caption pairs [11]. STPath, for instance, embeds each spot with a pathology foundation model before the transformer stage [2]. The ImageNet ResNet18 used in this paper is a deliberately simple and weaker choice, and we treat it as a limitation rather than a design goal.

D. Benchmarks and Data Quality

The benchmark in [7] evaluates eleven methods on accuracy, generalisability, translational potential, usability and computational efficiency, using several datasets and external validation. SpaRED [8] collects 26 public sources, evaluates eight prediction models, and notes that differences in datasets, preprocessing and training protocols make fair comparison difficult; it also addresses missing values caused by gene-capture dropout with a transformer-based completion model. Both works argue, in effect, that reported numbers are only as trustworthy as the protocol behind them.

TABLE I
COMPARATIVE ANALYSIS OF HISTOLOGY-TO-EXPRESSION STUDIES

Study	**Concept**	**Model and Data**	**Relation to This Work**
ST-Net [4]	Expression regression	CNN; breast-tumour ST	Supervised baseline
BLEEP [5]	Contrastive imputation	Bi-modal embedding; liver Visium	Alternative approach
OmiCLIP [1]	Visual–omics foundation model	CLIP-style; 2.2 M pairs	Inspiration only
STPath [2]	Slide-level prediction	Geometry-aware transformer	Inspiration only
STFlow [6]	Whole-slide generation	Flow matching	Related generative method
SPADE [9]	ST-informed latent space	Mixture of experts; HEST-1k	Related approach
UNI [10]	Pathology encoder	Self-supervised ViT	Future encoder
CONCH [11]	Vision–language encoder	CLIP-style model	Future encoder
Benchmark [7]	Method comparison	Five datasets and external validation	Strict evaluation
SpaRED [8]	Benchmark and completion	26 public sources	Fair-protocol motivation
This work	Patch, neighbours and position	ResNet18; ridge; fusion MLP	Leakage-aware skin baseline

E. Synthesis and Research Gap

Three observations follow from the literature. First, the field is moving toward scale, and the strongest reported systems rely on pathology-specific encoders and very large paired corpora [1]–[3], [10]. Second, evaluation practice is uneven: benchmarks [7], [8] show that preprocessing, dataset choice and splitting rules change the conclusions, and spatially adjacent spots in a random split can inflate scores. Third, within the studies reviewed here we did not find a small, fully inspectable baseline for inflammatory skin that holds out whole sections and reports every gene of a pre-declared panel. This paper addresses that gap. It does not try to outperform large pretrained models; it asks what a frozen general-purpose encoder, neighbourhood context and position can recover when whole sections are unseen, and which genes it cannot.

III. DATASET

A. GSE197023 Skin Visium Data

We use the public 10x Visium skin dataset GSE197023, generated for a study of the inflammatory cell network in

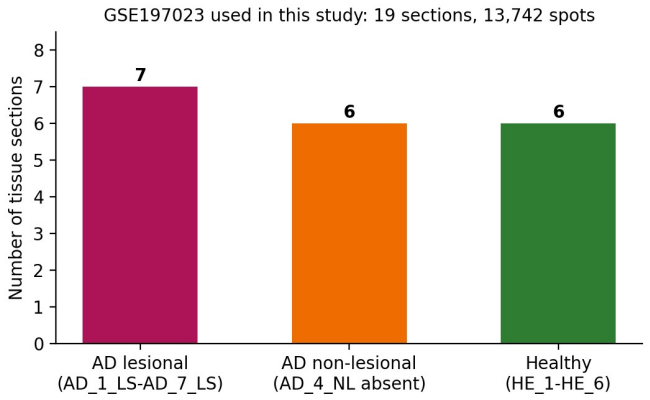


Fig. 2. Composition of the data used in this study: 19 sections in three groups.

atopic dermatitis [12]. Biopsies were taken from the upper arm of healthy donors and of AD patients, with lesional and non-lesional skin sampled from the patients [12]. Each Visium spot is about 55 µm across and therefore contains several cells. Table II lists the sections used here; sample AD_4_NL is absent from the files available to us, which leaves 19 sections.

TABLE II
SECTIONS USED IN THIS STUDY

Group	Sections	Identifiers
AD, lesional	7	AD_1_LS to AD_7_LS
AD, non-lesional	6	AD_1_NL, AD_2_NL, AD_3_NL, AD_5_NL, AD_6_NL, AD_7_NL
Healthy	6	HE_1 to HE_6
Total	19	13,742 spots with valid patches

B. Spots and Patches

A square 96 px patch was cropped around every spot from the high-resolution tissue image. Spots whose patch fell outside the image were removed, which leaves 13,742 spots with valid image patches. Image features are taken from the high-resolution tissue image and not from full-resolution whole-slide scans, and alignment between image and spots was checked visually for three sections (AD_2_LS, AD_2_NL and AD_6_LS).

IV. PROPOSED METHODOLOGY

A. System Architecture

The proposed system (Fig. 3) has five stages: data preprocessing, feature construction, gene-target construction, model training and sample-grouped evaluation. Each Visium section contributes an H&E image, a count matrix and spot coordinates. The image branch produces patch and neighbourhood features, the coordinate branch produces relative positions, and the count branch produces the 56-gene target matrix. Models map the three input blocks to the targets and are scored only on tissue sections that were held out as a whole.

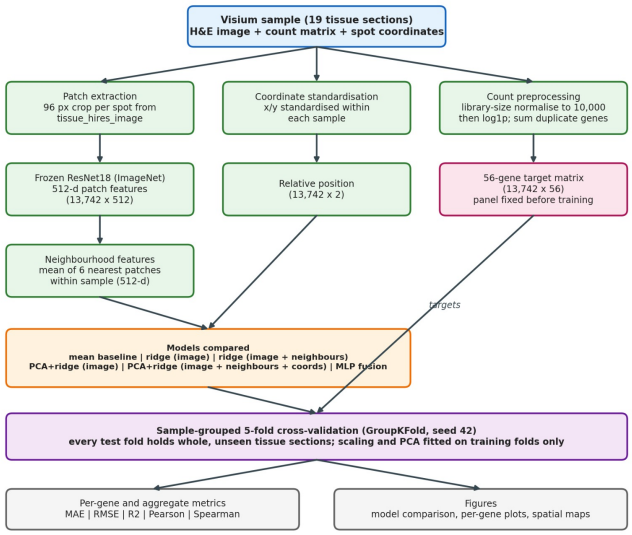


Fig. 3. Proposed pipeline. Image, position and count branches are processed separately and are joined only at the model stage; targets are used only for training and scoring. Diagram drawn by the authors.

B. Data Preprocessing

Raw counts are normalised per spot to a common library size of 10,000 and transformed with a logarithm. For gene j at spot i with count c_{ij}, the target is:

$$y_{ij} = \log\left(1 + 10{,}000 \cdot \frac{c_{ij}}{\sum_k c_{ik}}\right). \quad (1)$$

Duplicate gene symbols in the feature file are summed before normalisation. Pixel coordinates are not comparable across slides, so x and y are standardised within each section. Feature scaling, principal-component analysis and target scaling are fitted on the training folds only, so no statistic of a held-out section reaches the model.

C. Feature Engineering

Three input blocks are built for each spot. (i) Patch appearance: the 96 px patch is passed through a ResNet18 [13] with ImageNet weights, frozen, and the 512-dimensional pooled output is kept. (ii) Neighbourhood appearance: the mean of the ResNet18 features of the six nearest spots within the same section (512 dimensions), which gives the model local tissue context. (iii) Relative position: the two standardised coordinates. None of these blocks uses expression. A neighbour-averaged expression score exists in the repository as an exploratory diagnostic; it is derived from targets and is never used as a model input.

D. Gene Target Panel

The panel contains the 56 genes that are present in every section. It was fixed before model training, spans epidermal keratins, cornified-envelope, antimicrobial and inflammatory, immune, dermal and stromal, and vascular compartments, and no gene was removed or chosen on the basis of model performance. Detection rate, variance and Moran's I (a measure of

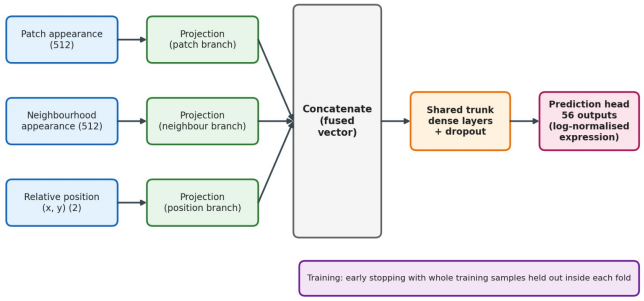


Fig. 4. Architecture of the multimodal fusion network. Diagram drawn by the authors.

spatial autocorrelation) are reported for description only and do not filter the panel.

E. Models

1) Baseline and Linear Models: The mean baseline predicts, for every gene, the mean of the training folds. Ridge regression is applied to image features only (ridge_image) and to image plus neighbourhood features (ridge_image_neighbor). Two further variants reduce the inputs to 128 principal components fitted on the training fold: pca_ridge_image, and pca_ridge_image_neighbor_coords, which also receives the two coordinates. Ridge penalties are selected with training data only. Linear models were chosen for their stability with 512 to 1,026 correlated inputs and a small number of sections, and because they expose how much each input block adds.

2) Multimodal Fusion Network: The fusion network (Fig. 4) gives each input block its own projection, concatenates the three projected vectors, and passes the result through a shared trunk with dropout to a head with 56 outputs. Separate projections prevent the 512-dimensional appearance blocks from overwhelming the two-dimensional position block. Early stopping is applied with whole training samples held out inside each fold, so the validation set used for stopping is also section-disjoint from the training data, and the test fold is never used for model selection.

F. Evaluation Protocol

The primary protocol is GroupKFold with 5 folds, grouped by tissue section, with random seed 42; every section is in a test fold exactly once. Leave-one-section-out (19 folds) is supported as an option. Random spot-level splits are not used. Neighbouring Visium spots share tissue context, so a random split would place closely related spots on both sides of the boundary and give optimistic estimates, a concern that benchmarking studies also raise about protocol differences [7], [8].

G. Evaluation Metrics

With y the measured and $\hat{y}$ the predicted value over N held-out spots, the pipeline reports:

$$\mathrm{MAE} = \frac{1}{N}\sum |y - \hat{y}| \tag{2}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum (y - \hat{y})^2} \tag{3}$$

$$R^2 = 1 - \frac{\sum (y - \hat{y})^2}{\sum (y - \bar{y})^2} \tag{4}$$

together with the Pearson and Spearman correlations between y and $\hat{y}$. A negative R^2 means that the model predicts that gene worse than a constant mean would. Within-sample Pearson correlation is computed inside individual sections and asks whether the spatial pattern within one section is recovered, as opposed to differences between sections. All metrics are reported per gene, per fold and in aggregate.

V. RESULTS AND OBSERVATIONS

All numbers in this section were produced automatically by the experiment script from out-of-fold predictions: every spot was predicted by a model that never saw its tissue section during training. The run covers 13,742 spots, 56 genes and 19 sections, with 5-fold sample-grouped cross-validation and seed 42.

A. Performance of All Models

The prediction-error and R^2 results are shown in Table III; correlation metrics are shown in Table IV.

TABLE III
MODEL COMPARISON: PREDICTION ERRORS AND R^2

Model	MAE	RMSE	Mean R^2	Positive R^2
mlp_fusion	0.7246	0.9478	0.0842	48/56
pca_ridge_image_neighbor_coords	0.7252	0.9521	0.0704	39/56
pca_ridge_image	0.7275	0.9518	0.0718	43/56
ridge_image	0.7298	0.9562	0.0624	40/56
ridge_image_neighbor	0.7377	0.9691	0.0366	33/56
mean_baseline	0.8251	1.0239	-0.0295	0/56

TABLE IV
MODEL COMPARISON: CORRELATION METRICS

Model	Mean Pearson	Within-Sample Pearson
mlp_fusion	0.2843	0.2158
pca_ridge_image_neighbor_coords	0.2821	0.2031
pca_ridge_image	0.2754	0.2012
ridge_image	0.2716	0.1987
ridge_image_neighbor	0.2696	0.1928
mean_baseline	-0.1997	N/A

Every image-based model improves on the training-mean baseline in MAE and RMSE. The fusion network is best on all reported metrics: its MAE of 0.7246 is about 12% lower than the 0.8251 of the baseline, and it is the only model with positive R^2 for 48 of the 56 genes. Its lead over the strongest linear model is small (MAE 0.7246 against 0.7252). With one seed and 19 sections we therefore describe the fusion network as consistently ahead but do not claim that the difference is statistically established.

The linear models show how the input blocks interact. Adding the 512 neighbourhood features to ridge regression lowered mean R^2 from 0.0624 to 0.0366 and the number of

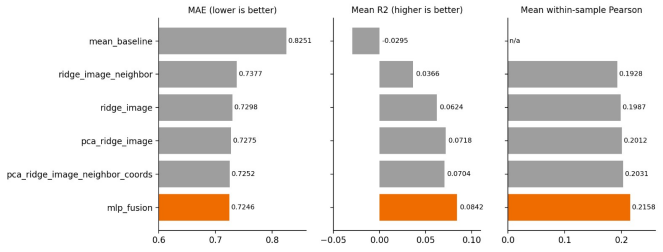


Fig. 5. Model comparison: MAE, mean R^2 and mean within-sample Pearson correlation (values from Tables III and IV). The MAE axis starts at 0.6. The within-sample correlation of the mean baseline is undefined because it predicts a constant.

positive genes from 40 to 33, which is consistent with over-parameterisation when the inputs double in size but the number of sections stays at 19. Reducing the inputs to 128 principal components raised mean R^2 to 0.0718 (43 positive genes). Adding neighbourhood features and coordinates to the PCA model gave the highest Pearson correlation among the linear models (0.2821) but a slightly lower mean R^2 (0.0704), so spatial context improved ranking more than calibration. The negative overall Pearson correlation of the baseline (-0.1997) is a property of fold-specific constant predictions and carries no information about image content.

B. Per-Gene Performance

Table V lists selected genes for the fusion network; the values for all 56 genes are available in RESULTS.md of the repository.

TABLE V
SELECTED PER-GENE RESULTS FOR MLP FUSION

Gene	MAE	RMSE	R^2
CASP14	0.1709	0.4205	0.4394
KRT1	0.1198	0.3854	0.4376
DCN	0.2197	0.4818	0.4340
KRT10	0.1014	0.3609	0.4282
DSG1	0.2210	0.4780	0.4279
CLDN1	0.1720	0.4183	0.4256
S100A8	0.2703	0.5373	0.1362
KRT6A	0.2283	0.5082	0.1003
S100A9	0.2064	0.4808	0.0882
CD8A	-0.0174	0.0434	0.0769
CXCL9	-0.0738	0.0392	0.0565
CXCL10	-0.0546	0.0616	0.0282

The highest within-sample correlations belong to CASP14, KRT1, DCN, KRT10, DSG1 and CLDN1 (0.43 to 0.44). The highest R^2 values belong to S100A8 (0.2703), KRT6A (0.2283), DSG1 (0.2210), DCN (0.2197) and SERPINB3 (0.2148). Eight genes have negative R^2 under the best model: LOR, DEFB4A, CD8A, CD68, CD4, CXCL9, LCN2 and CXCL10. They are reported unchanged.

C. Observations

1) Structure-Linked Genes Are Recovered Inside a Section: Differentiation and barrier genes of the epidermis (CASP14, KRT1, KRT10, DSG1, CLDN1) and the stromal gene DCN

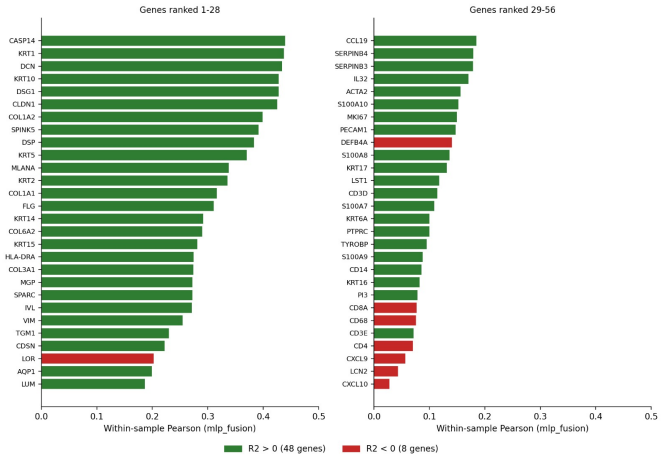


Fig. 6. Within-sample Pearson correlation of mlp_fusion for all 56 genes, ranked. Green: $R^2 > 0$ (48 genes); red: $R^2 < 0$ (8 genes).

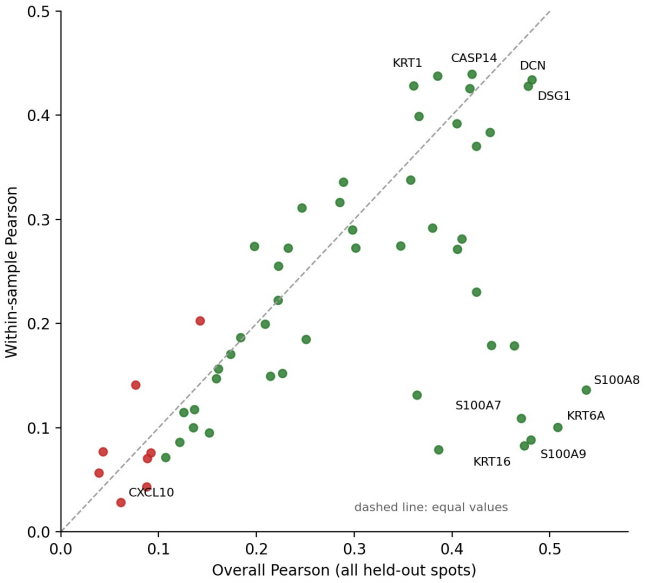


Fig. 7. Overall Pearson correlation against within-sample Pearson correlation for the 56 genes. Points below the dashed line are genes whose pooled correlation exceeds their within-section correlation.

have the best within-sample correlations. Epidermis and dermis are visually distinct in H&E, so a patch encoder can plausibly separate them even when it was not trained on pathology. This reading is exploratory.

2) Inflammatory Genes Are Separated Mainly Between Sections: S100A8, S100A9, KRT6A, KRT16 and S100A7 have overall Pearson correlations of 0.47 to 0.54 but within-sample correlations of only 0.08 to 0.14 (Fig. 7). Because the overall correlation pools spots from different sections, one explanation is that the model recognises disease-associated differences between sections, such as lesional against healthy tissue, better than the arrangement of expression within one section. The present metrics are consistent with this explanation but do not test it; a group-wise analysis of lesional, non-lesional and

healthy sections would.

3) Sparse Immune Genes Are Not Predictable Here: CD8A, CD4, CD3E, CXCL9 and CXCL10 have near-zero R^2 and low correlations. These transcripts are typically detected in few spots, and the repository's gene statistics (detection rate and variance) can be used to check this.

4) Practical Meaning: A mean R^2 of 0.0842 is small. The model carries some signal and ranks spots in a useful order for several genes, but it cannot replace a measured spatial transcriptomics experiment.

VI. CONCLUSION

A. Limitations

The evaluation uses 19 sections from one public dataset. Sample-grouped cross-validation controls leakage inside the dataset but is not external validation.

The encoder is a frozen ImageNet ResNet18, not a pathology-specific encoder, and it is not fine-tuned.

Only a fixed 56-gene panel is predicted, not the full transcriptome.

Visium spots contain several cells, so predictions are spot-level and not cell-level.

Features come from the high-resolution tissue image and not from full-resolution whole-slide images, and alignment was checked visually for three sections only.

Results come from one seed and one cross-validation setting, so small differences between models should not be over-interpreted.

Biological interpretation is exploratory. The work is not a clinical or diagnostic system.

B. Future Work

Replace the ResNet18 with a pathology encoder such as UNI [10] or CONCH [11].

Model the spot adjacency graph with graph neural networks.

Fine-tune the image backbone end-to-end.

Use full-resolution patches and multi-scale context.

Validate on an independent cohort, for example using HEST-1k [3].

Add calibration and uncertainty estimates for individual predictions.

C. Concluding Remarks

This paper presented a transparent and leakage-aware pipeline for predicting a 56-gene panel from H&E patches of human skin, using patch appearance, neighbourhood appearance and position. Under sample-grouped cross-validation, a multimodal fusion network gave the best results (MAE 0.7246, mean R^2 0.0842, 48 of 56 genes with positive R^2), and the largest gains were for genes tied to visible tissue structure. Inflammatory genes were separated mainly between sections, and sparse immune genes were not recoverable. The results show a modest but real morphology–expression signal for a frozen general-purpose encoder, and they mark where pathology encoders, larger cohorts and external validation would be needed next.

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Conflict of Interest— The authors declare that they have no known competing interests.

Data Availability— The dataset analysed in this study is the public 10x Visium skin dataset GSE197023 [12], available from the NCBI Gene Expression Omnibus. No new data were generated.

Code Availability— The code and generated result tables are available at github.com/manya0306/spatial-tissue-profiling.

Author Contributions— Manya Aggarwal: image processing and patch extraction, patch-quality and alignment checks, ResNet18 feature extraction, neighbourhood features, pipeline integration, and experiment and visualisation workflows. Aashna Das: gene-target construction, gene-level statistics, baseline, ridge, PCA and fusion modelling, sample-grouped evaluation design, metrics and per-gene analysis. Both authors contributed to problem formulation, literature review, experimental design, leakage prevention, analysis of limitations and the final manuscript.

REFERENCES

[1] W. Chen, P. Zhang, T. N. Tran, et al., "A visual–omics foundation model to bridge histopathology with spatial transcriptomics," Nature Methods, 2025, doi: 10.1038/s41592-025-02707-1.

[2] T. Huang et al., "STPath: A generative foundation model for integrating spatial transcriptomics and whole-slide images," npj Digital Medicine, vol. 8, Art. no. 659, 2025, doi: 10.1038/s41746-025-02020-3.

[3] G. Jaume, P. Doucet, A. H. Song, et al., "HEST-1k: A dataset for spatial transcriptomics and histology image analysis," in Advances in Neural Information Processing Systems, vol. 37, 2024, arXiv:2406.16192.

[4] B. He, L. Bergenstråhle, L. Stenbeck, et al., "Integrating spatial gene expression and breast tumour morphology via deep learning," Nature Biomedical Engineering, vol. 4, pp. 827–834, 2020.

[5] R. Xie, K. Pang, S. W. Chung, et al., "Spatially resolved gene expression prediction from histology images via bi-modal contrastive learning," in Advances in Neural Information Processing Systems, vol. 36, 2023, arXiv:2306.01859.

[6] T. Huang, T. Liu, M. Babadi, W. Jin, and R. Ying, "Scalable generation of spatial transcriptomics from histology images via whole-slide flow matching," in Proc. 42nd Int. Conf. on Machine Learning (ICML), 2025.

[7] C. Wang, A. S. Chan, X. Fu, S. Ghazanfar, J. Kim, E. Patrick, and J. Y. H. Yang, "Benchmarking the translational potential of spatial gene expression prediction from histology," Nature Communications, vol. 16, Art. no. 1544, 2025, doi: 10.1038/s41467-025-56618-y.

[8] D. Ruiz, P. Cárdenas, L. Manrique, D. Vega, G. Mejia, and P. Arbeláez, "Completing spatial transcriptomics data for gene expression prediction benchmarking," arXiv:2505.02980, 2025.

[9] E. Redekop, M. Pleasure, Z. Wang, K. Flores, A. Sisk, W. Speier, and C. W. Arnold, "SPADE: Spatial transcriptomics and pathology alignment using a mixture of data experts for an expressive latent space," arXiv:2506.21857, 2025.

[10] R. J. Chen, T. Ding, M. Y. Lu, et al., "Towards a general-purpose foundation model for computational pathology," Nature Medicine, vol. 30, no. 3, pp. 850–862, 2024, doi: 10.1038/s41591-024-02857-3.

[11] M. Y. Lu, B. Chen, D. F. K. Williamson, et al., "A visual-language foundation model for computational pathology," Nature Medicine, vol. 30, no. 3, pp. 863–874, 2024, doi: 10.1038/s41591-024-02856-4.

[12] Y. Mitamura, M. Reiger, J. Kim, et al., "Spatial transcriptomics combined with single-cell RNA-sequencing unravels the complex inflammatory cell network in atopic dermatitis," Allergy, vol. 78, no. 8, pp. 2215–2231, 2023, doi: 10.1111/all.15781.

[13] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), pp. 770–778, 2016.